

Module 06: Learning in Embodied

Learning Objectives

1. Define **Learning** within the context of Embodied.
2. Analyze how Learning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of Learning.

Introduction

This module explores **Learning**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Learning is a critical component of the 8-part Active Inference spine, bridging the gap between Action and Communication.

Key Concepts

1. Learning as a Markov Blanket Boundary

How does Learning define the boundary between the agent and the environment?

2. Generative Models of Learning

What parameters involved in Learning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Learning drive the perception-action loop?

Applications

In Embodied, we see Learning manifest in: * **Specific Example 1:** A medical student initially needs to consciously check each diagnostic criterion, but after years of clinical residency, the

pattern recognition becomes automatic -- she walks into a room and "just knows" the diagnosis from the patient's gait, skin color, and the quality of their voice; this transition from effortful reasoning to intuitive knowing is the generative model undergoing structure learning, compressing explicit rules into embodied priors that operate as fast, low-free-energy pattern completion. * **Specific Example 2:** A potter learning to throw on the wheel initially struggles with conscious coordination of hand pressure, wheel speed, and clay moisture, but after thousands of pots, her hands intuitively feel the exact moment to pull upward and the precise pressure to thin the wall; this learning trajectory shows the generative model progressively absorbing sensorimotor regularities until prediction errors during throwing approach zero, and what was once effortful computation becomes immediate bodily knowing.

Conclusion

Understanding Learning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.